

+ 92/100

8.2

13. $c_1=c_2=2; c_3=c_4=c_5=3$

$$A = \begin{bmatrix} -1 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & -1 & 1 \\ -1 & 0 & 1 & 0 \end{bmatrix} \quad C = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}$$

With Python:

$$A = \begin{bmatrix} -2 & 2 & 0 & 0 \\ 0 & -2 & 2 & 0 \\ 0 & 0 & -3 & 3 \\ -3 & 0 & 0 & -3 \end{bmatrix} \quad A^T = \begin{bmatrix} -1 & 0 & 0 & 1 & -1 \\ 1 & -1 & 0 & 0 & 0 \\ 0 & 1 & -1 & 0 & 1 \\ 0 & 0 & 1 & -1 & 0 \end{bmatrix}$$

$$A^T C A = \begin{bmatrix} 8 & -2 & -3 & -3 \\ -2 & 4 & -2 & 0 \\ -3 & -2 & 6 & -3 \\ -3 & 0 & -3 & 6 \end{bmatrix}$$

Solving it gives $x = \begin{bmatrix} 1 \\ 3 \\ 1 \\ 1 \\ 0 \end{bmatrix}$

$$y = -(Ax) = \begin{bmatrix} 1 \\ 2 \\ 1 \\ 2 \\ -1 \end{bmatrix}$$

17. a. With nullspace $\Rightarrow \text{rank}(A) = n-1 = 8$

b. Since $N(A) = \{c_1, c_2, \dots, c_8\}$

$1^T f = 0 \Rightarrow f_1 + f_2 + \dots + f_8 = 0$

$\Rightarrow \text{KCL}$

c. $\sum_{i=1}^8 2 = 2(12) = 24$ or $\text{trace}(A^T A) = 24$

4.1

7. $x_1 - x_2 = 1$ Solve $A^T y = 0; y^T b = 1$

$x_2 - x_3 = 1$
 $x_3 - x_1 = 1$
 $A^T = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 2 & 4 \\ 2 & 3 & 5 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = 0$

$= \begin{bmatrix} 1 & 2 & 3 \\ 0 & -2 & -1 \\ 0 & -1 & -1 \end{bmatrix}$

$\Rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 0 & -1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \Rightarrow y_2 + y_3 = 0 \Rightarrow y_2 = -y_3$

$y_1 + 2y_2 + 3y_3 = 0$

$\Rightarrow y_1 - 2 + 3 = 0 \Rightarrow y_1 = -1$

$\therefore y = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$

9. - Column Space

- Orthogonal

More specific

32. all y all must be orthogonal to u and have dimensions

$n + (n-r) = n; r + (m-r) = m$
 $(C(A) \perp N(A)) \quad (C(A^T) \perp N(A^T))$

b. $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ $\leftarrow$ quick, be general

31. $N = \text{null}(A)$

$N^T = \text{null}(A^T)$

$B = \text{null}(\text{null}(A^T))$

$N^T = m-r$ components

"Cancel out" the null $\Rightarrow$ Row space

33. a. when those vectors

are orthogonal and

satisfy: $r + (n-r) = n$

$r + (m-r) = m$

b. $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

4.2

13. $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad b = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}$

$\hat{p} = A \hat{x} = A(A^T A)^{-1} A^T b$

$(A^T A)^{-1} = \left(\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right)^{-1}$

$= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

$A(A^T A)^{-1} A^T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = A$

$Ab = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix} \Rightarrow AAb = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}$

Shape: $m=4, n=1$

16. $a \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} + b \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \approx \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}$

$\Rightarrow \begin{bmatrix} 1 & 1 \\ 2 & 0 \\ 3 & 1 \end{bmatrix} \hat{x} \approx \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}$

$A^T A \hat{x} = A^T b$

$A^T A = \begin{bmatrix} 1 & 2 \\ 2 & 2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix}$

$A^T b = \begin{bmatrix} 1 & 2 \\ 2 & 2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 6 \\ 10 \\ 5 \end{bmatrix}$

$\Rightarrow \hat{x} = \begin{bmatrix} \frac{1}{2} \\ \frac{3}{2} \end{bmatrix}$

17. $P^2 = P \Rightarrow (I-P)^2 = I-P$

$\Rightarrow (I-P)^2 = I^2 - 2IP + P^2$

$\Rightarrow I - 2IP - P = I - 2P + P$

$= I - P$

work explanation: $P = \text{Nullspace}(A)$

30. a. $(A) = \begin{bmatrix} 3 \\ 4 \end{bmatrix} = A \quad A^T = \begin{bmatrix} 3 & 4 \end{bmatrix}$

$P = A(A^T A)^{-1} A^T = \begin{bmatrix} 3 \\ 4 \end{bmatrix} \begin{bmatrix} 3 & 4 \end{bmatrix}^{-1} \begin{bmatrix} 3 & 4 \end{bmatrix}$

b. $(A^T) = \begin{bmatrix} 3 \\ 4 \end{bmatrix} = A^T$

$P = A(A^T A)^{-1} A^T$

$A^T = \begin{bmatrix} 1 & 2 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 \\ 2 \end{bmatrix} \begin{bmatrix} 1 & 2 \end{bmatrix}^{-1} \begin{bmatrix} 1 & 2 \end{bmatrix}$

$= \frac{1}{5} \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 1 & 2 \end{bmatrix}$

$P_c A P_r = A \Rightarrow P^2 = P$

b. checks and check $e = b - p$ is orthogonal to A .